Panskura Banamali College (Autonomous)

B.Sc.(H) 3rd Semester Examination 2022 (Online mode)

Subject : Computer Science Paper: CC5T

Paper Name : Data structures

F.M. : 40 Time : 2 Hours

Answer any four questions **4×10=40**

1. a) What is complexity of an algorithm? How do you push elements in a linked stack .
   b) What is threaded binary tree ? What is the difference between tree and graph ?
   (2+3)+(2+3)
2. a) What is sparse matrix . Explain with example.
   b) Difference between priority queue and De queue.
   (2+3)+5
3. a) Write an algorithm which accepts a stack as input and reverse the element of the stack using one additional queue.
   b) Prove that E=I+2q where E is the external path length , I is the internal path length and q is the number of vertices that are not leaves in a tree.
   5+5
4. a) Construct a binary search tree from the given values .
   Consider the 1st value as the root value.
   Values – 45,23,29,85,92,7,11,35,49,51
   b)Prove that a linked binary tree with n nodes ,n ≥0 has exactly (n+1) NULL links.
   5+5
5. a) Difference between complete binary tree and almost complete binary tree.
   b) Write an algorithm to evaluate the given postfix expression .Using this algorithm to find the value of
   7,3,4,+,-,4,9,3,/,+,*,3,^,4,+
   5+5
6. Write a c function to explain quick sort. Explain the time complexity of merge sort.5+5

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